UNDEGRADUATE STUDIES RESEARCH TEMPLATE INSTRUCTIONS (MODIFIED TRADITIONAL FORMAT)

Guide to Manuscript Preparation and Organization

These guidelines outline the formatting and structural requirements for submitting a manuscript.

1. Title Page Requirements

The initial page must serve as a comprehensive overview of the submission and include the following details:

- **Manuscript Title:** A precise, descriptive title limited to a maximum of two lines. Use sentence-case capitalization (only capitalizing the first word, proper nouns, and scientific genus names). Do not include abbreviations or numerical values.
- **Short Title:** A brief running head (e.g., *Heavy metals and Biosorption*) that captures the core topic. It must not exceed 70–80 characters, including spaces.
- **Author Names:** Present names in the order of First Name/Initials, Middle Name/Initials, and Surname. Ensure correct spelling, as names will be published exactly as submitted. Avoid using all-caps.

2. Abstract Page

To preserve a double-blind peer-review process, the abstract page must contain only the title, the abstract text, and keywords. No author names or institutional identifiers should appear from this page onward.

- **Abstract Text:** Provide a standalone, accessible, and informative introduction to the research. It must be written as a single paragraph without internal headings, though it should conceptually flow through four areas: Background, Methodology, Results (Key findings only), and Conclusion.
    - **Length:** Maximum of 350 words.
    - **Restrictions:** Do not include citations or specialized jargon/abbreviations.
    - **Closing:** The final sentence must emphasize the broader impact and relevance of the study.
- **Keywords:** Include 5 to 8 indexing terms, arranged in alphabetical order, that accurately reflect the manuscript's subject matter.

3. Main Text Structure

The core text should generally follow the standard IMRAD format: Introduction, Materials and Methods, Results, and Discussion, Conclusion and Recommendations.

Introduction

- Define the core objectives, hypotheses, or research questions.
- Provide necessary background context and a balanced literature review. Acknowledge conflicting findings in the field rather than selecting only supportive data.
- Keep this section highly specific, well-referenced, and accessible to non-specialists.
- **Note:** Do not preview the study's results or conclusions here.

Materials and Methods

- Provide enough procedural detail to allow complete replication of the experiments.
- Established techniques can be cited rather than described in full; however, unique or novel protocols require exhaustive detail.
- **Animal Research:** Must comply with the most recent ARRIVE guidelines and explicitly state approval from an institutional animal care and use committee.
- **Human Subjects:** Must adhere to STROBE (or equivalent) guidelines and explicitly state institutional review board (IRB) approval.

Results and Discussion

Authors may choose to present each section separately, or merge these two sections into a single, cohesive unit.

- **Results:** Present data chronologically and logically, supported by relevant statistical significance testing. Use the past tense.
    - Report all relevant findings, not just statistically significant or hypothesis-supporting data.
    - Avoid superfluous experimental details and do not introduce methods for the first time here.
    - Indicate the preferred placement of figures and tables in the left margin of the text. Never duplicate data across both a graph and a table. Subheadings may be used.
- **Discussion:** Interpret the findings and highlight their primary implications concisely.
    - Contextualize the findings using up-to-date literature without over-extrapolating beyond what the data proves.
    - Offer a transparent evaluation of the study's limitations. Do not introduce new methods or results here.

Conclusion

- The **Conclusion** section provides a holistic, definitive statement that synthesizes the entire study rather than merely listing individual results. Structurally, it begins by answering the initial research question before broadening in scope to outline real-world implications and future research avenues, which are best presented in a bulleted number format. Finally, to preserve scientific integrity, this section must remain highly focused and entirely free of new empirical data, unmentioned arguments, or external literature citations.

Recommendations

The **Recommendations** section of a research paper translates empirical findings into specific, actionable steps for both practical application and future scientific inquiry. Structurally, it provides concrete advice for industry practitioners and policymakers on how to implement the study's data to solve real-world problems, optimize workflows, or update standards. Concurrently, it establishes a strategic roadmap for academia by explicitly directing future researchers on how to scale the work, explore unexamined variables, and overcome the study's identified methodological limitations. To maintain scientific integrity, every recommendation must be realistic, highly specific rather than generalized, and derived exclusively from the verified data presented within the manuscript.

4. References

- **Style:** Follow the "Vancouver" referencing system outlined by the International Committee of Medical Journal Editors (ICMJE).
- **In-Text Citation:** Use non-superscripted numbers in square brackets (e.g., [1], [3–5]). Order citations sequentially based on their appearance in the text, tables, or figures.
- **Tables:** Include citations in a dedicated column using square-bracketed numbers.
- **Restrictions:** Each number must correspond to only one reference. Do not cite unpublished data or papers.

5. Visuals and Data Presentation

Figures

- **Formatting:** Submit high-resolution color graphics on a white background using American English spelling. Do not include grid lines on charts.
- **Submission:** Save each figure as an individual file (TIFF or JPG preferred), named using the corresponding author's surname and the figure number (e.g., Anderson Figure 1a). If tables contain embedded images, provide separate high-resolution files for them.
- **Sizing:** Create original images between 3.5 and 4.5 inches (8.9–11.4 cm) wide to prevent distortion. Peer review allows lower resolutions (>90 dpi), but final manuscript requires high-resolution files (300 dpi; max width 9 inches, max height 6 inches).
- **Captions:** Embed figures of Data results, with appropriate captions, within the main body text at the Results section. Explicitly credit external image sources at the end of the caption (e.g., *"Courtesy of SK Smith"*).
- **Structure:** Number figures sequentially with Arabic numerals (Figure 1, Figure 2) in the order they are referenced.

Tables

- **Creation:** Generate tables exclusively using the native table tools in Microsoft Word or Libre/OpenOffice.
- **Placement:** Embed tables within the main body text at the Results section.
- **Structure:** Split oversized data into separate tables. Number tables sequentially with Arabic numerals (Table 1, Table 2) in the order they are referenced.
- **Captions & Footnotes:** Provide a descriptive title for each table. If data is pulled from a prior publication, cite the source at the end of the caption. Use lowercase italicized superscript letters for general footnotes and asterisks for statistical significance metrics. Place footnotes directly below the table body; do not put footnotes in the table title.

Submission Formatting Requirements

Authors must strictly align their submissions with the formatting criteria outlined below. Please be advised that the editorial staff cannot perform basic formatting adjustments for authors. Submissions that do not comply with these structural standards will be returned to the sender without undergoing peer review.

1. Document Properties and Layout

- **File Formats:** Submit manuscripts exclusively as editable, un-password-protected .docx files.
- **Font Selection:** Use 12-point font in either *Times New Roman* or *Calibri*.
- **Spacing and Margins:** Format the entire document with double-spacing and generous margins on both sides.
- **Navigation:** Enable continuous page numbering (positioned at the bottom right) and continuous line numbers throughout the entire file.
- **Indentation and Automation:** Use native software commands or tabs for paragraphs and indents rather than manual spacing. Do not use field functions.
- **Structural Layout:** Never embed visual data within the main body of the text. Position all tables and figures sequentially at the very end of the file, following the bibliography.
- **Paper Size:** Use short bond paper size (21.59 x 27.94 cm)
- **Page Numbers.** Place page numbers on lower right corner of the page.

2. Article Type Restrictions and Content Distributions

1. Target Length

For Research Articles under Traditional format, no target length is required, but authors must balance comprehensive detail with concise synthesis.

2. Structural Distribution (The "IMRAD" Model)

2.1 Title Page & Abstract

- **Purpose:** To frame the study and offer a scannable summary.
- **Execution:** The abstract must be a single, dense paragraph capped at 350 words. It acts as a microcosm of the paper, detailing the backdrop, methods, core metrics, and overarching conclusion. It is followed by 5 to 8 alphabetized keywords.

2.2 Introduction

- **Purpose:** To establish the "why" behind the research.
- **Execution:** Rather than providing an exhaustive historical review, focus strictly on the immediate context of the problem. Define the knowledge gap, present a balanced overview of the current literature (including conflicting data), and clearly state the study's central hypothesis or primary objectives.

2.3 Materials and Methods

- **Purpose:** to provide a transparent, step-by-step blueprint of how the research was conducted so that other scholars can validate the approach, trust the results, and replicate the study

- **Execution:** This section is highly technical and written in the future tense (for proposal papers) and past tense (for post-data collection research defense output). Standard, established protocols should be cited rather than written out in full. However, any novel methodologies or specialized modifications require exhaustive, step-by-step detail.
- *Compliance Note:* Research involving animals or human subjects must explicitly state adherence to international guidelines (e.g., ARRIVE or STROBE) and note institutional review board (IRB) approval codes.

2.4 Results

- **Purpose:** To report data objectively and transparently without interpretation or discussion.
- **Execution:** Present findings in a logical, chronological sequence that directly mirrors the narrative and order of the Methodology.
- **Data Integrity:** Include all outcomes relevant to the hypothesis, regardless of whether they are statistically significant or favorable. Avoid descriptive redundancy—do not repeat specific data points in the text if they are already clearly presented in a table or figure.

2.5 Discussion

- **Purpose:** To interpret the findings, explain what they mean, and contextualize their broader scientific impact.
- **Execution:** Construct a tightly argued narrative that relates your outcomes back to the existing literature introduced in the Introduction. Authors must also provide a transparent, objective evaluation of the study's limitations.

2.6 Conclusions & Recommendations

- **Purpose:** To provide a holistic synthesis of the study and offer actionable steps for future work.
- **Execution:** Answer the central research question based on your findings and highlight real-world implications and future research directions—ideally summarized in a clear, bulleted or numbered format.
- **Integrity Guardrail:** Strictly avoid introducing any new data, analysis, or external literature in this section to preserve scientific integrity.

OR

2.4 Results & Discussion

- *Purpose: To report data objectively while simultaneously interpreting the findings and contextualizing their broader impact.*
- *Execution: Present findings in a logical, chronological sequence that mirrors the narrative of the Methodology. Construct a tightly argued narrative by immediately relating your outcomes back to the relevant literature introduced in the Introduction.*
- *Data Integrity & Context: Include all outcomes relevant to the hypothesis, not just statistically significant or favorable results. Avoid descriptive redundancy—do not repeat data*

points in the text if they are already clearly laid out in a table or figure. Authors must also provide a transparent evaluation of the study's limitations within this narrative.

2.5 Conclusions & Recommendations

- *Purpose: To provide a holistic synthesis of the study and offer actionable steps for future work.*
- *Execution: Answer the central research question based on your findings, highlighting real-world implications and future research directions—ideally summarized in a clear, bulleted or numbered format.*
- *Integrity Guardrail: Strictly avoid introducing any new data, analysis, or external literature in this section to preserve scientific integrity.*

3. Language, Terminology, and Data Style

- **Language:** Submissions must be written entirely in English.
- **Headings:** Organize the paper using a maximum of three structural levels for headings and subheadings.
- **Nomenclature and Metrics:** Apply universally recognized scientific nomenclature. All measurements must use standard symbols conforming to the International System of Units (SI).
- **Mathematical Material:** Embed equations using an editable equation editor tool. Do not insert mathematical statements as static image files.

4. Handling Acronyms and Abbreviations

- **Standard Terms:** Use globally accepted scientific and professional acronyms.
- **Initial Definition:** Spell out all abbreviations completely upon their first mention in the text, placing the short form in parentheses immediately following—for example: *Polychloride biphenyl (PCB)*.
- **Non-Standard Terms:** Avoid uncommon abbreviations unless they are used a minimum of three times in the text and are defined at their first introduction.
- **Inventory:** Compile a comprehensive index of all non-standard abbreviations at the conclusion of the main text, directly preceding the Acknowledgments section.

5. Page & Visual Layout Strategy

- **Formatting Specifications:** The document must be double-spaced using a standard 12-point font (Times New Roman or Calibri) with wide margins. Continuous page and line numbering are required to facilitate the review process.

- **Table Restrictions:** Oversized datasets should be partitioned into separate, smaller tables.

- **Visual Submissions:** Figures must be high-resolution (300 dpi for final publication) on a solid white background. Captions are compiled on a dedicated page separate from the images themselves.

- **Placement Rule:** Embed figures or tables directly inside the body text.

6. Back Matter and Front Matter Checklist

Ensure the following short sections are included to satisfy transparency and ethical standards:

1. **Table of Contents.** The **Table of Contents (TOC)** section of a research paper serves as a structured, sequential roadmap that outlines the document's organizational hierarchy using titles, headings, and subheadings paired with right-aligned page numbers. It systematically inventories the manuscript's layout across three primary divisions: the front matter (preliminary pages like acknowledgments, abstracts, and lists of visuals, often indexed with lowercase Roman numerals), the core chapters (the main body sections adhering to the IMRAD framework), and the back matter (supplementary resources including the bibliography and appendices). To ensure formal navigation and clarity, a well-executed TOC uses progressive indentation to denote heading depth, incorporates dot leaders to guide the reader's eye, and maintains flawless verbal and pagination alignment with the actual text inside the document. Place this section as the page right after Abstract.
2. **List of Visuals.** The **List of Visuals** section—frequently separated into a consecutive **List of Tables** and **List of Figures**—serves as a formal directory within the front matter of a research paper to help readers instantly locate non-textual data assets. Structurally, it catalogs matrix-based data arrays under the "Table" designation and all charts, graphs, diagrams, and maps under the "Figure" designation, arranging them sequentially using either a continuous or chapter-based numbering system. Each entry in the list features the asset's exact number, its verbatim title or primary descriptive caption left-aligned, a horizontal row of dot leaders, and the precise page number right-aligned. Placed immediately following the Table of Contents, this section maintains absolute verbal symmetry with the text and is paginated using lowercase Roman numerals to denote its status as preliminary metadata.
3. **Acknowledgements:** Explicit recognition of institutional funding, grant numbers, or logistical assistance from non-authors (requires their prior consent), and other important personalities deemed necessary for acknowledgement by the research authors. Place this section as the page right after List of Visuals.
4. **Conflict of Interest Statement:** A mandatory disclosure of any financial or personal ties that could be perceived as biasing the work. Place this section at the Appendices.
5. **Bibliography:** A sequential, numbered reference list formatted strictly to Vancouver style, corresponding directly to the square-bracketed citations used throughout the text.
6. **Plagiarism checking results.** The maximum tolerable plagiarism result, as per Grammarly Pro scanning, in the Institution is set at 10%. Anything above this limit is advised to revise the problematic contents of the paper. Place the screenshot of the first page of the official plagiarism scanning result at the Appendices.
7. **Other Sections in the Appendices.** The Researchers may opt to add more sections in the Appendices such as photo documentations of the actual sampling of the study, letters to dignitaries as proof of official communications, Data Collection forms, etc.

Chronological Flow of the Research Paper

FRONT MATTER

1. Title Page

2. Abstract
3. Table of Contents
4. List of Visuals
5. Acknowledgements

6. MAIN TEXT (Traditional 5-Chapter Format)

CHAPTER 1: INTRODUCTION

- Background of the Study
- Research Literature (related local and international studies)
- Statement of the Problem
- Research Objectives (General & Specific)
- Research Hypothesis (if applicable)
- Significance of the Study
- Scope and Limitations
- Definition of Terms

CHAPTER 2: METHODOLOGY **or** MATERIALS AND METHODS

- Research Design
- Study Locale
- Respondents and Sampling Technique (inclusion/exclusion criteria)
- Research Instrument
- Data Collection Procedure
- Ethical Considerations

• Data Analysis Plan

CHAPTER 3: RESULTS AND DISCUSSION **OR** RESULTS, then DISCUSSION

• Objective presentation of findings (no interpretation)

• Organized by specific research objectives

• Supported by tables, figures, and statistical results

• Followed by analysis and interpretation contextualized with related literature

CHAPTER 4: CONCLUSION, AND RECOMMENDATIONS, **OR** CONCLUSION, then RECOMMENDATIONS

• Summary of Findings

• Conclusions (direct answers to the research problem)

• Recommendations for practice, policy, and future research

BACK MATTER

1. References

◦ Formatted using the Vancouver style

◦ Numbered sequentially as cited in the text

2. Appendices

◦ Appendix A: Research Instrument / Data Collection Form

◦ Appendix B: Approval Letters / Ethical Clearance

◦ Appendix C: Statistical Computations / Raw Data Summary

◦ Appendix D: Curriculum Vitae of Researchers

◦ Appendix E: Plagiarism Scanning Result

◦ Appendix F: Other Supporting Documents